RESEARCH PROPOSAL · TITLE

Adaptive Context Reasoning System (ACRS)

A Structural Orchestration Layer for Multi-Agent LLM Ecosystems

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Proposal stage. No empirical performance claims. Live Indian Open Government Data is a methodological constraint, not a result.

RESEARCH PROPOSAL · RESEARCH STUDENT DETAILS

Research student details

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RESEARCH PROPOSAL · CONTENTS

Contents

Nine required sections. Research Methodology is expanded for each objective (O1 SATR, O2 APRR, O3 MNCD, O4 FCNP) plus the integrated SATR → APRR → MNCD → FCNP loop. Objective order is SATR → APRR → MNCD → FCNP (not SMART).

RESEARCH PROPOSAL · INTRODUCTION

Introduction

This proposal treats multi-agent LLM systems as a computer-science systems problem: not a new foundation model, but a missing orchestration layer between session memory, specialist routing, live tools, and context growth.

Wu et al. introduce AutoGen as a conversation-driven programming framework in which agents exchange messages until a stopping condition (ICLR 2024 LLM Agents Workshop; COLM 2024, arXiv:2308.08155). Hong et al. encode Standard Operating Procedures into MetaGPT so that a software-company metaphor produces structured artefacts (ICLR 2024). Qian et al. organise ChatDev as a chat-chain of organisational roles (ACL 2024). Li et al. study communicative agents in CAMEL (NeurIPS 2023).

Tool use is a parallel line. Qin et al. release ToolLLM / ToolBench: 16k+ REST APIs, a DFSDT planner, and ToolEval (ICLR 2024). Zheng et al. add ToolRerank over ToolLLM candidates (LREC-COLING 2024). Learned routers (RouteLLM, MasRouter, PILOT) pick models or collaboration modes. LLMLingua shortens prompts by token importance (EMNLP 2023).

Those stacks still leave four operational surfaces underspecified as one contract: session–tool fusion, a training-free specialist posterior, fail-loud live Indian Open Government Data, and a mesh prune that writes a residue back into retrieval.

ACRS is proposed as that missing layer. The integration order is SATR → APRR → MNCD → FCNP. The live demonstration corpus is Agriculture on data.gov.in. This deck does not claim a leaderboard number.

RESEARCH PROPOSAL · LITERATURE REVIEW

Literature Review

Multi-agent systems literature. What the cited papers actually contribute, and what they leave open for a structural orchestration layer.

Wu et al., AutoGen (COLM 2024 / ICLR 2024 workshop): conversation as the programming model. Gap: messages, not (toolId, score, live citation), are the unit of coordination.

Hong et al., MetaGPT (ICLR 2024): authored SOPs reduce role drift. Gap: the graph of who speaks next is written by the designer, not updated from session-local affinity after a live tool call.

Qian et al., ChatDev (ACL 2024): organisational chat-chain for software artefacts. Gap: the environment is a codebase, not a ministry API.

Li et al., CAMEL (NeurIPS 2023): inception prompting and communicative role-play. Gap: no first-class vote over tool identifiers backed by Open Government Data.

Taken together, these systems prove conversation, SOPs, software roles, and inception prompting. None of them is a session–route–mesh–prune loop over live Indian OGD.

RESEARCH PROPOSAL · LITERATURE REVIEW

Literature Review — tools, routing, and compression

Tool learning, learned routers, and prompt compression are real literatures. They are not substitutes for the four ACRS modules.

Qin et al., ToolLLM / ToolBench (ICLR 2024): SBERT API retriever, DFSDT planner, ToolEval. Zheng et al., ToolRerank (LREC-COLING 2024): contrastive rerank of ToolLLM candidates. Gap: both are turn-amnesic; they do not fuse which tools actually fired into the next rank.

Ong et al., RouteLLM (2024); Yue et al., MasRouter (ACL 2025); Panda et al., PILOT (Findings of EMNLP 2025): learned or bandit routers over LLM SKUs or collaboration modes. Gap: they do not maintain a training-free Dirichlet–Thompson posterior over named tool specialists.

Jiang et al., LLMLingua (EMNLP 2023): token-importance prompt compression. Gap: compression does not prune a mesh by conductance or write a residue back into retrieval.

Guo, Woodruff & Yadav, PECAD (AAAI 2020): AGMARKNET as a decision-support input for price prediction. Gap: a crop-yield / price CNN is not a multi-agent live-OGD loop. Cited as domain precedent, not as a baseline to beat.

Summary of Literature Review

Qualitative map only. This table names the gap each family leaves for ACRS; it does not claim a percentage improvement over any baseline.

Literature family	What it already does	What ACRS still has to add
MAS frameworks (AutoGen, MetaGPT, ChatDev, CAMEL)	Conversation, SOPs, software roles, inception prompting	A closed session–route–mesh–prune contract, not another chat runtime
Tool learning (ToolLLM / ToolBench / ToolRerank)	Large-scale tool-use corpus and ToolEval-style ranking protocol	Session-fused ranking plus live Indian OGD execution (not RapidAPI replay)
Learned routers (RouteLLM, MasRouter, PILOT)	Train or bandit-route over model SKUs or collaboration modes	Training-free Dirichlet–Thompson posterior over named specialists
Prompt compression (LLMLingua)	Shorten tokens before the LLM	Prune a mesh by conductance and write the residue back into SATR
Indian agriculture DSS (PECAD)	Shows AGMARKNET as a real decision-support input	Live, fail-loud OGD inside a multi-agent loop — not a crop-yield model to beat

Identified Research Problem

Five architectural gaps. Four named objectives. One integration contract. Closing a gap is evidenced by a runnable loop and citable equations, not by a promised leaderboard number.

> G1 — Turn-amnesic retrieval. ToolLLM SBERT (Qin et al., ICLR 2024) and ToolRerank (Zheng et al., LREC-COLING 2024) score each query independently. Session co-activation is unused. → Objective 1 SATR.

> G2 — Routers pick models or authored SOPs, not tool-specialist agents with a training-free posterior. RouteLLM / PILOT / MasRouter / MetaGPT. → Objective 2 APRR.

> G3 — MAS communication is star, SOP, or chat. No mesh vote whose object is a live tool identifier. AutoGen, MetaGPT, ChatDev, CAMEL. → Objective 3 MNCD.

> G4 — Prompt compressors (LLMLingua, EMNLP 2023) do not write live citations back into retrieval. → Objective 4 FCNP.

> G5 — No closed four-stage contract executed on journal-publishable Indian OGD (data.gov.in / AGMARKNET / IMD / DES), with ToolBench used only as a ranking library. → Integrated ACRS.

Gaps are architectural. This deck does not convert them into NDCG, latency, or accuracy targets.

Research Title & Aim

Adaptive Context Reasoning System (ACRS): A Structural Orchestration Layer for Multi-Agent LLM Ecosystems

> Aim. Design, implement, and critically evaluate ACRS — a structural orchestration layer in which session-aware tool retrieval (SATR), training-free specialist routing (APRR), mesh consensus over tool identifiers (MNCD), and flow-coupled context pruning with write-back (FCNP) form a closed loop on live Indian Open Government Data. The proposal-stage claim is architectural completeness and live-pipeline integrity, not a leaderboard number.

> Scope of the title. Adaptive = session-conditioned ranking and routing. Context = fused session memory plus mesh residue. Reasoning = specialist posterior plus score-sum consensus. System = four named modules on one contract.

> What the title is not. It is not a new LLM, not a crop-yield model, and not a claim that ACRS already outperforms ToolLLM, MasRouter, or LLMLingua on a published leaderboard.

> Domain lock for the live loop. Agriculture on data.gov.in (AGMARKNET and related verified UUIDs). Other sectors stay out of the inference path in this proposal.

> Integration order that the aim implies. SATR (memory) → APRR (who acts) → MNCD (live evidence) → FCNP (prune and write back).

Research Objectives

Four named modules. Each row is a thesis-sized design claim. Metric targets are deferred until a protocol is frozen.

ID	Objective	What will be designed
O1	SessionRerank+ (SATR)	The retrieval chapter of the thesis: given query q and session history H, return a ranked list of ToolBench-schema tools fused with a co-activation cache. The unit of publication is the fusion rule, not an NDCG target.
O2	Adaptive Probabilistic Routing Reinforcement (APRR)	The routing chapter: a training-free posterior over tool-specialist agents, updated from SATR scores and from live MNCD observations. The unit of publication is the Bayesian controller, not a Pareto chart.
O3	Mesh Network Context Diffusion (MNCD)	The consensus-and-execution chapter: gossip (toolId, score), score-sum consensus, live data.gov.in GET on verified UUIDs. The unit of publication is the protocol plus the citation contract.
O4	Flow-Coupled Network Pruning (FCNP)	The memory chapter: Kirchhoff/Physarum conductances prune the mesh trace and write surviving live citations back into SATR. The unit of publication is the coupling, not a token-percentage.

RESEARCH PROPOSAL · RESEARCH OBJECTIVES

Objective 1 — SATR

The retrieval chapter of the thesis: given query q and session history H, return a ranked list of ToolBench-schema tools fused with a co-activation cache. The unit of publication is the fusion rule, not an NDCG target.

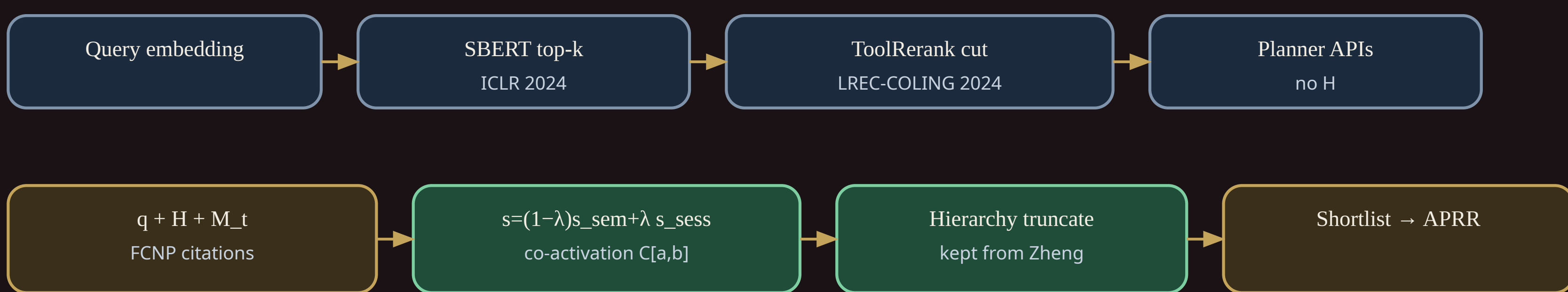
SOTA. Qin et al., ToolLLM / ToolBench (ICLR 2024): Sentence-BERT API retriever over 16,464 RapidAPI tools, then ToolLLaMA + DFSDT. Zheng et al., ToolRerank (LREC-COLING 2024): adaptive truncation of seen vs unseen APIs and hierarchy-aware concentration/diversity. Pipeline: instruction → SBERT retrieve top-k APIs → (optional ToolRerank truncate/rerank) → LLM DFSDT/ReAct planner

Gap. SOTA retrievers are turn-amnesic. They do not maintain a success-conditioned co-activation graph or ingest pruned memory from later stages.

Novelty. Session co-activation cache as a first-class prior over tool pairs. Convex fusion of semantic and session scores; λ may grow with session length. ToolRerank-style seen/unseen truncation kept, then applied after session scoring. FCNP memory mixed into the next SATR prior so retrieval is closed-loop.

O1 comparison — SOTA retrieval vs SATR (no NDCG drawn)

Top: Qin ToolLLM + Zheng ToolRerank (turn-amnesic). Bottom: session fusion + co-activation.



Repository: github.com/joyjeni/session-aware-toolbench-rerank

RESEARCH PROPOSAL · RESEARCH OBJECTIVES

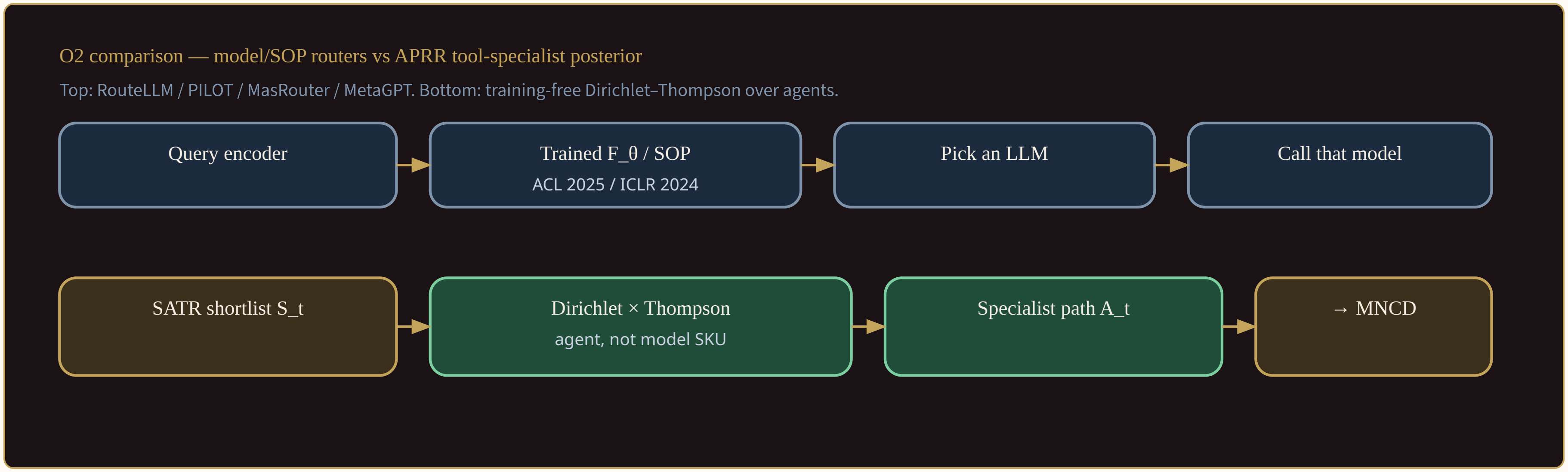
Objective 2 — APRR

The routing chapter: a training-free posterior over tool-specialist agents, updated from SATR scores and from live MNCD observations. The unit of publication is the Bayesian controller, not a Pareto chart.

SOTA. Yue et al., MasRouter (ACL 2025, doi:10.18653/v1/2025.acl-long.757): trained neural controller over multi-agent topologies. Ong et al., RouteLLM (arXiv:2406.18665, 2024): routers among LLMs. Panda et al., PILOT (Findings of EMNLP 2025): preference-prior LinUCB for budget-constrained LLM routing. Hong et al., MetaGPT (ICLR 2024): authored SOP workflows. Pipeline: query → trained controller or difficulty model → choose one LLM/agent → execute

Gap. SOTA either trains a neural router, picks a model, or follows an authored SOP. It does not maintain a training-free affinity matrix over Indian OGD tool families.

Novelty. Training-free online W versus MasRouter's learned controller. Implemented update is $\kappa \cdot reward \cdot 1/L^2 \cdot 1/lat\_norm$ with negative reward on failure. CTGR and FTDR remain in the GitHub repo; this lab runs core APRR hops. W is session state, so routing can adapt across farmer turns.



Repository: github.com/joyjeni/aprr-multi-agent-routing

Objective 3 — MNCD

The consensus-and-execution chapter: gossip (toolId, score), score-sum consensus, live data.gov.in GET on verified UUIDs. The unit of publication is the protocol plus the citation contract.

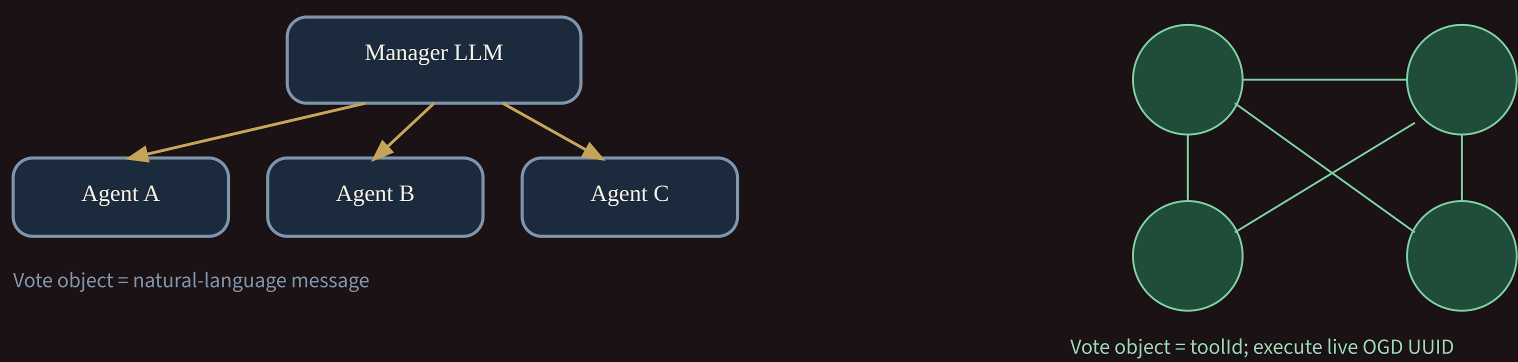
SOTA. Wu et al., AutoGen (ICLR 2024 LLM Agents Workshop; COLM 2024, arXiv:2308.08155): multi-agent conversation. Hong et al., MetaGPT (ICLR 2024): SOP pipeline. Qian et al., ChatDev (ACL 2024): organisational chat-chain. Li et al., CAMEL (NeurIPS 2023): communicative role-playing agents. Pipeline: manager LLM → sequential or star-topology agent messages → tool calls

Gap. Star and chat topologies coordinate over natural-language messages. They do not vote over tool identifiers backed by a ministry API.

Novelty. First-class vote object is a tool ID, not a chat utterance. consensus_pick is score-sum; consensus_pick_borda is a non-default variant. Live Indian OGD only. Catalog-only tools stay ranking-only; prices are never invented. HTTP 5xx/429 are retried; empty filters show other live rows from the same resource.

O3 comparison — star/chat SOTA vs mesh vote over live tool IDs

Left: AutoGen / MetaGPT / ChatDev / CAMEL. Right: MNCD gossip + score-sum + data.gov.in.



Repository: github.com/joyjeni/mncd-mesh-agents

RESEARCH PROPOSAL · RESEARCH OBJECTIVES

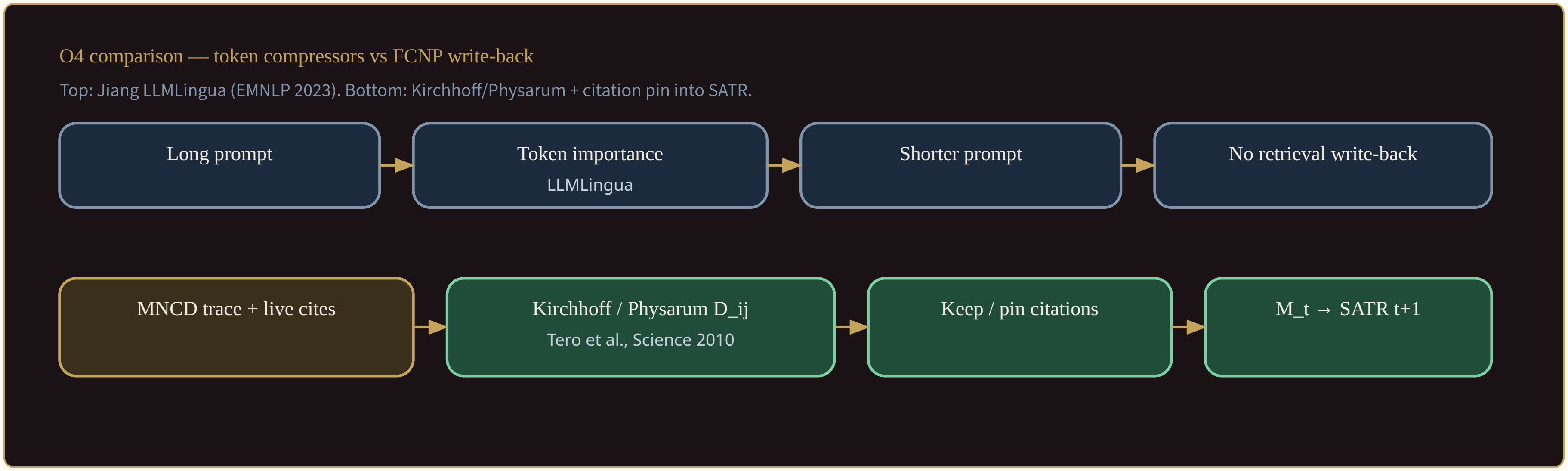
Objective 4 — FCNP

The memory chapter: Kirchhoff/Physarum conductances prune the mesh trace and write surviving live citations back into SATR. The unit of publication is the coupling, not a token-percentage.

SOTA. Jiang et al., LLMLingua (EMNLP 2023): token-level prompt compression. Tero et al., Science 2010 (doi:10.1126/science.1177894): Physarum adaptive network. Park et al., Generative Agents (CHI 2023): language memory stream in a sandbox. Pipeline: long prompt → compressor → LLM. Memory is not written back into a tool retriever.

Gap. Token compressors are not current-reinforced over a context graph and do not pin live government citations into the next retrieval turn.

Novelty. Laplacian solve with grounded sink, matching the published Python pruner. Hybrid tiering and persistent high-flow citations. If a requested crop/state has no AGMARKNET rows today, other live rows are shown; nothing is invented. Closed loop: retained spans become SATR session memory.



Repository: github.com/joyjeni/fcnp-context-pruning

RESEARCH PROPOSAL · RESEARCH QUESTIONS

Research Questions

Each question is paired with one objective. Answers will be empirical after a protocol is frozen; this slide does not pre-commit scores.

> RQ1. Can tool retrieval be conditioned on a co-activation cache and session memory rather than a single query embedding (Qin et al., ToolLLM, ICLR 2024; Zheng et al., ToolRerank, LREC-COLING 2024)? → addressed by O1 (SessionRerank+ (SATR)).

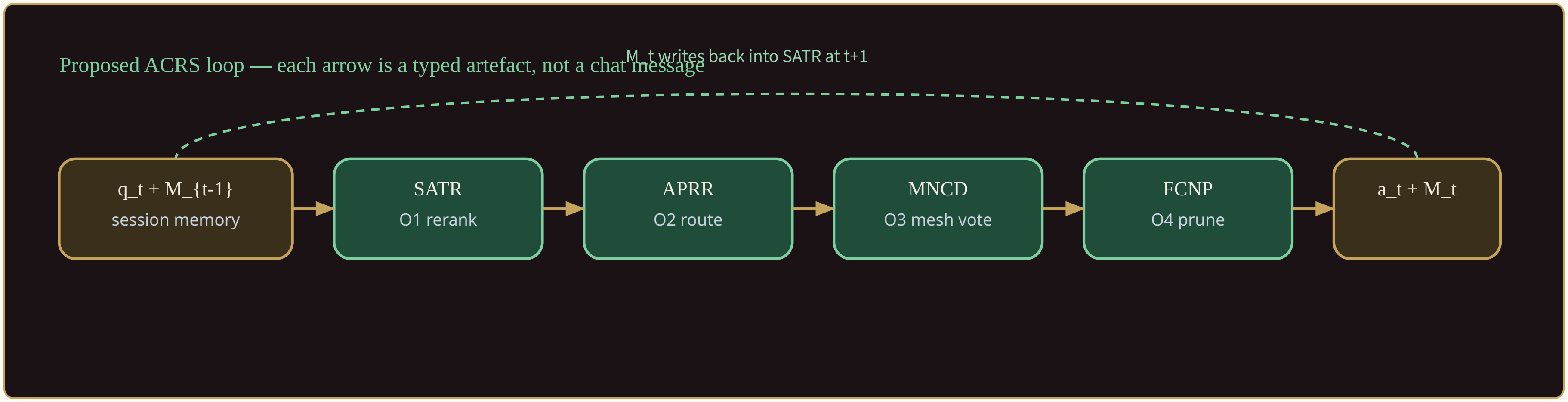
> RQ2. Can routing sample a training-free posterior over tool-specialist agents instead of a trained neural controller or an authored SOP (Yue et al., MasRouter, ACL 2025; Hong et al., MetaGPT, ICLR 2024; Ong et al., RouteLLM, 2024)? → addressed by O2 (Adaptive Probabilistic Routing Reinforcement (APRR)).

> RQ3. Can execution proceed as gossiped (toolId, score) plus score-sum consensus, then a live data.gov.in GET, instead of a central chat orchestrator (Wu et al., AutoGen, COLM 2024)? → addressed by O3 (Mesh Network Context Diffusion (MNCD)).

> RQ4. Can Kirchhoff/Physarum pruning pin live citations back into retrieval rather than only shortening the prompt (Jiang et al., LLMLingua, EMNLP 2023; Tero et al., Science 2010)? → addressed by O4 (Flow-Coupled Network Pruning (FCNP)).

Research Methodology

One methodology per objective, then one integrated protocol. Proposal-stage: design the loop and freeze the protocol; do not pre-commit a leaderboard.



- O1 SATR — fused session ranking against a ToolBench-style library. Live mandi rows are not SATR's job.
- O2 APRR — training-free Dirichlet–Thompson posterior over named specialists. Learned routers are a future bake-off, not the implementation.
- O3 MNCD — gossip + score-sum consensus, then live data.gov.in GET on verified Agriculture UUIDs. Fail loud. No dummy prices.
- O4 FCNP — Kirchhoff / Physarum-inspired conductance prune with write-back into SATR. Not LLMLingua token deletion.
- Integrated contract. SATR → APRR → MNCD → FCNP must run in that order on one user turn. Skipping MNCD or FCNP is an incomplete run.
- What will not appear in this proposal. Promised retrieval scores, latency targets, consensus percentages, or token-reduction ratios.

RESEARCH PROPOSAL · RESEARCH METHODOLOGY

Research Methodology — O1 SATR

How session-aware ranking will be designed and later evaluated.

- Design method. Specify a fused session object: query + dialogue turns + last tool traces + co-activation counts. Rank tools with that object, not with the raw utterance alone.
- Ranking library (not live prices). Use ToolBench / ToolLLM artefacts as a public tool-ranking library and protocol family. RapidAPI-style traces are ranking evidence only.
- Live Indian OGD is out of SATR's ranking path. Mandi and weather rows enter at MNCD. SATR must not invent or cache dummy AGMARKNET prices.
- Implementation path. Persist co-activation in a session store; expose a rank(query, session) API that APRR can call. Fail loud if the session schema is incomplete.
- Future evaluation protocol (not a result). When the protocol is frozen, compare session-fused ranking against query-only ranking on the same ToolBench-style split. Report the protocol, not a pre-committed score.
- Deliverable for the thesis chapter. Algorithm, schema, and ablation plan (with vs without tool-trace fusion).

RESEARCH PROPOSAL · RESEARCH METHODOLOGY

Research Methodology — O2 APRR

How the specialist posterior will be designed. No win-rate or millisecond target is claimed at proposal stage.

Design method. Maintain a Dirichlet–Thompson posterior over named specialists. Sample or take the MAP specialist given SATR's fused session.

Allocation rule (proposal form). $P(\text{agent} \mid \text{context}) \propto W^\alpha \cdot \eta^\beta \cdot \psi^\gamma$, with W = session-matched specialist weight, η = reliability, ψ = cost/risk. Exponents are design knobs, not fitted claims.

Training-free stance. Do not train a MasRouter- or RouteLLM-style classifier as the primary method. Learned routers remain a future bake-off class, not the implementation.

What is being routed. Specialists in the ACRS mesh — not LLM SKUs (GPT-4 vs Mixtral) and not AutoGen conversation modes as the object of routing.

Update rule. After MNCD returns verified or failed evidence, update η (and optionally W) so the next turn's posterior is not identical to a cold start.

Future evaluation protocol (not a result). Log specialist choice vs task type on held-out session traces; compare against a static role graph. No pre-committed accuracy.

Research Methodology — O3 MNCD

How live Indian OGD and score-sum consensus will be executed. PECAD is a domain precedent, not a baseline to beat.

> Design method. Specialists selected by APRR gossip partial beliefs. Aggregate with score-sum (not Borda). Every live claim must cite a verified data.gov.in resource UUID.

> Corpus / API. Agriculture-only Open Government Data: AGMARKNET (9ef84268-d588-465a-a308-a864a43d0070) and other UUIDs that pass verification.

> Fail-loud contract. HTTP 5xx/429 are retried with backoff; missing API key, unverified UUID, or empty filtered universe hard-fail. No dummy mandi prices.

> Platform constraint. data.gov.in Elastic max_result_window is 10 000; limit is capped. Pagination is sequential, not a fabricated parallel harvest.

> Related work used correctly. Guo, Woodruff & Yadav, PECAD (AAAI 2020) shows AGMARKNET as a real DSS input. MNCD does not re-implement crop-yield prediction.

> Future evaluation protocol (not a result). Measure citation completeness, fail-loud rate on injected faults, and qualitative agreement of score-sum vs majority vote.

RESEARCH PROPOSAL · RESEARCH METHODOLOGY

Research Methodology — O4 FCNP

How mesh pruning will be designed. No token-reduction ratio is claimed at proposal stage.

> Design method. Treat the post-MNCD mesh as a flow network. Apply a Kirchhoff / Physarum-inspired conductance update; drop low-conductance specialist edges; keep a residue.

> Write-back (the integration hinge). The residue is written into SATR's next fused session. Without write-back, FCNP would be prompt compression by another name.

> Contrast with LLMLingua. LLMLingua shortens tokens before the LLM. FCNP prunes who remains in the mesh. Tokenisers are not the primary artefact.

> Heuristic honesty. Discrete conductance is a design heuristic inspired by Tero et al. (Science, 2010), not a proof that the mesh is a Physarum organism.

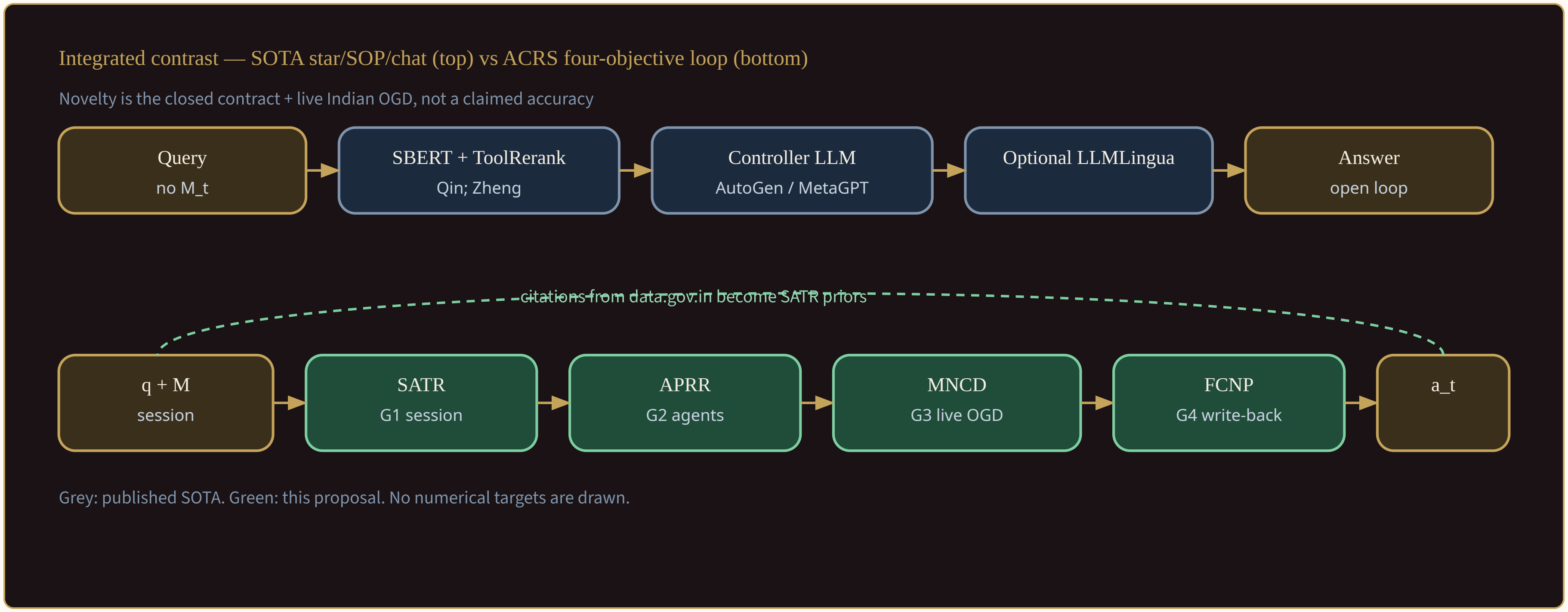
> Safety. Pruning must not delete the last live-data specialist if MNCD still has an open verified query. Fail loud rather than silently drop evidence.

> Future evaluation protocol (not a result). Compare mesh size and downstream SATR rank stability with vs without pruning on the same session traces.

RESEARCH PROPOSAL · RESEARCH METHODOLOGY

Research Methodology — integrated loop

Closed contract SATR → APRR → MNCD → FCNP → SATR. The novelty is the loop and the data contract, not a claimed accuracy.



Closed-loop protocol. One user turn must traverse all four modules in order. Skipping MNCD (no live UUID) or FCNP (no write-back) is treated as an incomplete run, not a successful demo.

Two evidence regimes. (1) ToolBench-style ranking traces for SATR. (2) Live data.gov.in Agriculture rows for MNCD. Do not mix dummy prices into (1) or RapidAPI tools into (2).

Independent variables (future experiments). Session fusion on/off; Dirichlet routing vs static roles; score-sum vs majority; FCNP write-back on/off.

Dependent measures (to be frozen later). Citation completeness, fail-loud correctness, rank stability across turns, qualitative specialist-choice logs.

Domain lock. Agriculture on Indian OGD for the live path. Other sectors are out of scope until a later amendment.

Ethics / data. Public government catalogues only; no personal data; API keys stay in the environment, never in the thesis text.

Master repository: github.com/joyjeni/phd-agentic-ai-master

Conclusion

ACRS is proposed as a structural orchestration layer. The contribution is the closed loop and the four named gaps — not a pre-committed leaderboard.

> The literature (AutoGen, MetaGPT, ChatDev, CAMEL, ToolLLM, MasRouter, RouteLLM, PILOT, LLMLingua, PECAD) establishes conversation, tools, learned routing, and prompt compression. It does not establish SATR → APRR → MNCD → FCNP as one contract over live Indian OGD.

> The identified problem is five gaps: G1 session–tool fusion, G2 training-free specialist posterior, G3 live fail-loud Indian OGD, G4 conductance prune with write-back, G5 missing integration layer.

> The aim is to design that layer. Four objectives (SATR, APRR, MNCD, FCNP) and four research questions map onto it.

> Methodology is specified per objective: ToolBench as ranking library; Dirichlet–Thompson routing; verified data.gov.in UUIDs with score-sum; Physarum-inspired prune with SATR write-back. Metric numbers are deferred.

> Next step after approval. Freeze evaluation protocols, implement the closed loop, and report whatever the measurements show — including negative results.

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CrewAI is an engineering framework without a flagship peer-reviewed paper in this list; AutoGen and MetaGPT are the MAS citations.

RESEARCH PROPOSAL · THANK YOU

Thank you

Adaptive Context Reasoning System — a research programme, not a performance number.

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